An epidemiological model of monkeypox: model prediction and control application

> Author

Abstract

Background Monkeypox (Mpox) is an emerging infectious disease caused by the Mpox virus (MPX Virus). The outbreak of Mpox epidemic has caused global panic and is now a public health incident. Various approaches have been proposed in the recent literature to study and analyze the epidemiological dynamics of this infection and effective prevention and control measures. Using mathematical model to understand the transmission dynamics and control strategy is a useful way to understand the prevention of Mpox.

Methods A new compartment model is established to examine the effectiveness of vaccine on Mpox based on previous studies. Nonlinear least squares fitting is used for model's parameter estimation. The impact of a series of preventive and control measures on the epidemic control is explored with optimal control theory in conjunction with the official data released by the authorities.

Results Firstly, a stability analysis of the developed model was carried out to show that, under certain circumstances, its equilibrium is both locally and globally stable. Secondly, based on the reported cases of Mpox infection in the United States between 2022 and 2023, the model's optimal parameter values were obtained. A sensitivity analysis of the model parameters was then conducted to identify the key parameters that affect the development of Mpox epidemics in the United States. Lastly, the comparison of control effects under various control strategies showed that implementing the all suggested four control measures at the same time was the most effective way to curb the development of monkeypox epidemic in the United States.

Conclusions This study has theoretical significance for understanding and controlling Mpox virus transmission.

Keywords Mpox transmission model, Parameter estimation, Sensitivity analysis, Optimal control strategy, United State

Introduction

In the long history of mankind, infectious diseases have always been accompanied by human beings. Up to now, there have been countless epidemic virus attacks on human beings, such as plague, measles, cholera, smallpox, tuberculosis, black death, AIDS, Middle East respiratory syndrome, Ebola virus, COVID- 19 and so on. The fight against the pandemic virus is arduous and lengthy, and will always pose threats and obstacles to human life, health and social development to varying degrees. At a time when the world has just passed the threshold of the outbreak of the COVID- 19, another outbreak of the global

Mpox epidemic will undoubtedly deal a heavy blow to the world's health resources.

Mpox, an infectious disease caused by Mpox virus (MPXV) [1], is a novel zoonotic disease. Not only can it spread rapidly between people, but it can also be transmitted to humans through contact with infected animals.

Mpox virus was first discovered in 1959, when there were two outbreaks of non-fatal variola disease in a population of cynomorphic monkeys from Singapore received in Copenhagen, Denmark. Magnus et al. isolated Mpox virus from infected animals, hence the name Mpox [2, 3]. A new Mpox outbreak began in May 2022. The first case of Mpox was reported in the United Kingdom on 7 May 2022, and an individual with a history of travel to Nigeria was confirmed [4, 5]. On May 15, the Centers for Disease Control and Prevention (CDC) reported a case of Mpox in Massachusetts [6], and since then, the Mpox virus has spread around the world at an unexpected rate. Within a few months, several countries, including the United States, were hit by the Mpox virus, and reported Mpox cases surged, forcing the World Health Organization (WHO) to declare the disease a Public Health Emergency of International Concern (PHEIC) on July 21, 2022 [7], causing widespread global public health concern. Outbreak data reported by the WHO shows that since the start of the Mpox outbreak, 117 countries have reported 93,921 confirmed cases of infection as of January 2024.

Animal to human transmission, human to human transmission and environmental transmission are possible and potential transmission routes of Mpox virus. Transmission of the virus may occur when humans come into physical contact with wild animals such as rodents and primates infected with Mpox virus (including being bitten or scratched, or coming into direct contact with the bodily fluids of infected animals during activities such as hunting, skilling, trapping, and cooking) [8]. Personto-person transmission is mainly carried out through close contact with infected persons, including face to face for a long time (such as talking or breathing close to each other to contact respiratory droplets of infected persons), skin contact (such as touching or kissing damaged skin to contact body fluids of infected persons), mouth to mouth (such as kissing) and other behaviors [8]. Compared to past Mpox outbreaks, this outbreak presents a unique epidemiological feature in that the virus is transmitted primarily through sexual contact, and in particular the unprecedented number of Mpox cases transmitted by sexual contact between men [9–11]. Mpox virus may also live for some time on the surfaces of objects such as clothing, towels, bedding, electronics that have been touched by patients, and environmental transmission can occur when people or animals interact with these items. Especially if the contact has skin lesions or touches organs or mucous membranes such as eyes, nose and mouth without washing hands, it may increase the risk of infection.

According to WHO, Mpox virus is a orthopoxvirus, which belongs to the genus Orthopoxvirus of poxviridae with smallpox virus [12]. Human infection with Mpox virus causes symptoms similar to, but milder than, smallpox, and the severity varies from person to person. The incubation period for Mpox is

generally 7–14 days, but may be as long as 21 days [13]. After the incubation period, prodrome symptoms begin to appear, usually including fever, chills, headache, muscle aches, fatigue and gland (lymph node) swelling, etc. Lymph node enlargement is a significant feature of Mpox and the main basis for distinguishing Mpox infection from infection of other members of the poxvirus family [7]. The signature rash (which looks like a blister or ulcer) will appear about 0–5 days later, and the rash will last about 2–4 weeks. In most cases, Mpox is self-limiting, meaning symptoms resolve on their own within a few weeks. However, immunocompromised groups, such as infants, children, pregnant women and people with underlying immune deficiencies, as well as people with advanced HIV, may experience more severe symptoms or complications and even death. According to available data, the mortality rate for Mpox is between 0.1% and 10%, with the majority of deaths occurring in young infected people [14].

There is currently no licensed vaccine and specific treatment for Mpox virus infection, but Mpox infection can be prevented and treated through vaccination against smallpox and antiviral drugs to prevent smallpox. Smallpox vaccine has been shown to be cross-protective against Mpox, providing about 85% protection, but currently only about 25% of people endemic to Mpox are vaccinated against smallpox [15–17]. Since the"Enhanced Smallpox Eradication Program"successfully achieved global eradication of smallpox in the 1980 s, smallpox vaccination has been discontinued, and people's immunity to the virus has declined significantly. In particular, people under the age of 40 lack immunity to the Mpox virus and are more susceptible to infection, which may also be the reason for the significant increase in the incidence of Mpox in the past decade [18]. In November 2022, WHO issued the Interim Guidelines for Mpox Vaccine and Immunization report, suggesting that the recommended vaccines mainly include ACAM2000, MVA-BN and LC16[19]. Currently, there is limited clinical data on the efficacy of these vaccines against Mpox, and it is not possible to know their exact effectiveness and safety. However, during the study of clinical efficacy, some people who received the above vaccine were still diagnosed with Mpox virus, indicating that the effectiveness of the vaccine was not 100% (in fact, the effectiveness of one dose of MVA-BN vaccine was only 35.8%) [20]. Therefore, on the one hand, relevant departments should pay more attention to improving the effectiveness and safety of the drugs and vaccines currently used for the prevention and treatment of Mpox virus, and at the same time, more manpower and material resources should be invested in the research and development of specific therapeutic drugs for Mpox virus, so as to provide more practical and effective guarantee for the prevention and control of Mpox epidemic.

The infectious disease dynamics model plays an important role in the study of infectious diseases and is widely used to describe the transmission dynamics of various diseases [21]. Since the global outbreak of Mpox in 2022, it has aroused wide attention from all walks of life and further promoted the study of Mpox dynamic model. In 2011, Bhunu and Mushayabasa [22] first tried to analyze the transmission dynamics of Mpox in human and non-human interactions using SIR dynamics model, solved the disease-free equilibrium point and endemic

equilibrium point of the model, and proved its stability. Olaniyi and Chuma [23] studied a monkeypox model with vertical transmission, which considered vaccination categories and optimal control interventions as well as a comprehensive cost–benefit analysis. Allehiany et al. [24] investigated the dynamics of monkeypox outbreaks in the United States using real statistical data, established mathematical models, performed parameter estimation and sensitivity analysis, and gave corresponding recommendations for the control and management of monkeypox. Omame et al. [25] considered that men who have sex with men are thought to be a new route of transmission for recent monkeypox outbreaks, and this population is also affected by HIV infection. Therefore, by developing a mathematical model of the mechanism of HIV and monkeypox virus co-infection, they explored the potential impact of HIV on monkeypox outbreaks in men who have sex with men. The analysis showed that HIV can promote monkeypox transmission among men who have sex with men. Recently, Abidem et al. [26] proposed a new Mpox dynamic chamber model focusing on quarantine levels and studied the effects of various control parameters on Mpox infection, exposure and isolation.

In order to study how to reduce the number of infected people with the smallest cost, so as to propose the best prevention and treatment strategies, many experts combine the epidemiological model with the optimal control theory [27, 28]. Samreen et al. [29] used a mature optimization theory to study Mpox disease dynamics under the influence of several control strategies for constant and transient variables, and proposed optimal control interventions accordingly. Peter et al. [30] established and analyzed an optimal control model with four control variables to study the dynamics of Mpox, showing that implementing prevention strategies for transmission from rodents to humans is the most economical and effective. Yap Hushenkan et al. [31] proposed a new Mpox transmission model, which introduced control variables to simulate the optimal prevention and treatment strategies. Numerical simulation showed that the proposed combined control strategy was necessary to prevent the Mpox epidemic.

Considering the protective efficacy of the monkeypox vaccine, that is, people vaccinated with the monkeypox vaccine are not completely immune to the monkeypox virus and are still at risk of infection after exposure to the monkeypox virus, we wanted to explore the impact of vaccine protective efficacy on disease transmission. Based on the traditional SEIR model of infectious disease dynamics, a vaccine compartment was added, and a seven-compartment differential equation system was constructed to describe the interaction between humans and non-humans by introducing a modified factor to represent the change in Mpox virus infection rate after vaccination. After a brief theoretical analysis of the established model, the development trend of Mpox epidemic in the United States during the period of 2022–2023 was explored using the data of Mpox infection cases in the United States, and suggestions were provided for epidemic prevention and control.

The main structure and contents of this paper are as follows: Section 2 introduces the process of establishing the model in detail and gives a kind of

Mpox dynamic model considering the effect of vaccination which is mainly analyzed and studied in this paper. In Section 3, a simple qualitative analysis of the model is carried out, including the global stability of the effective reproduction number, disease-free and endemic equilibrium points under certain conditions; In Section 4, the parameter estimation and sensitivity analysis of the model are mainly carried out, and the specific values of the key parameters in the model are obtained by using nonlinear least square method, and the sensitivity analysis of the model parameters to the effective reproduction number is given. In Section 5, the optimal control theory is applied to the original model, a new optimal control model is established, and the most appropriate prevention and control measures are sought under the optimal control framework through numerical simulation. In Section 6, numerical simulation is used to verify the validity of theoretical analysis. In Section 7, the main research contents and conclusions of this paper are summarized and prospected.

Model description

Given that Mpox is a zoonotic disease with the potential to be transmitted both between human populations and from animal hosts to humans, the model incorporates two distinct populations: the human population and the non-human (rodent) population. The total human population (N_h) is classified into five distinct segments: Susceptible humans (S_h), Vaccinated humans (V_h), Exposed humans (E_h), Infected humans (I_h), and Recovered humans (R_h). The non-human population is denoted by N_r. Given the current limited understanding of the dynamics of Mpox virus transmission among rodents, only two rodent subpopulations are considered: Susceptible rodents (S_r) and Infected rodents (I_r).

We give the following hypothesis: Mpox virus can be transmitted by human contact, as well as from rodent to human. Susceptible rodents can be infected by infectious rodents, but not by infectious humans. Rodents are still contagious and there is no recovery class. Susceptible humans can be transferred to vaccinated human populations (V_h) by vaccination, but vaccines reduce the risk of infection and do not provide 100% protection. Infected humans (I_h) transfer to the recovered human population (R_h) after recovery and gain immunity to the virus. All humans and rodents are evenly mixed in the same area. The total human population (N_h) and rodent population (N_r) are constant.

Therefore, based on the above descriptions, the corresponding mathematical equations are given by a system of non-linear ordinary differential equations below: Where $\lambda_{\mathrm{h}} = \frac{\beta_1 I_h + \beta_2 I_r}{N_h}$, $\lambda_r = \frac{\beta_3 I_r}{N_r}$.

Figure 1 gives the flowchart corresponding to the developed model (2.1), and the description of the parameters included in the model are given in Table 1.

Model analysis

Positivity and boundedness of the solutions

$$\text{Let } \mathrm{R}_7^+ = \left\{ (x_1, x_2, x_3, x_4, x_5, x_6, x_7)^T : x_i \geq 0, \forall 1 \leq i \leq 7 \right\}$$

Theorem 3.1. *Let the initial values of the state variables of Mpox disease dynamical system (2.1) be given as.*

$S_h(0) \geq 0,\ V_h(0) \geq 0,\ E_h(0) \geq 0,\ I_h(0) \geq 0,\ R_h(0) \geq 0, S_r(0) \geq 0, I_r(0) \geq 0.$ Then the solutions $\Omega = (S_h, V_h, E_h, I_h, R_h, S_r, I_r) \in \mathrm{R}_+^7$ of (2.1) are non-negative for all time $t > 0$.

The proof is given in Appendix A.

The boundedness of the solutions also be proved using the following theorem:

Theorem 3.2. *For any solution set* $(S_h, V_h, E_h, I_h, R_h, S_r, I_r) \in R_+^7$ *of system (*2.1*), there exists a feasible domain* $\Omega = \Omega_h \times \Omega_r$ *containing it, where* Ω_h *and* Ω_r *represent the feasible domain of the human and rodent subsystem, respectively.*

$$\begin{aligned}
\frac{dS_h(t)}{dt} &= \varphi_h - \lambda_h S_h - (\mu_h + \alpha_h) S_h, \\
\frac{dV_h(t)}{dt} &= \alpha_h S_h - \omega \lambda_h V_h - \mu_h V_h, \\
\frac{dE_h(t)}{dt} &= \lambda_h S_h + \omega \lambda_h V_h - (\varepsilon + \mu_h) E_h, \\
\frac{dI_h(t)}{dt} &= \varepsilon E_h - (\mu_h + \rho_h + \delta_h) I_h, \\
\frac{dR_h(t)}{dt} &= \rho_h I_h - \mu_h R_h, \\
\frac{dS_r(t)}{dt} &= \varphi_r - \lambda_r S_r - \mu_r S_r, \\
\frac{dI_r(t)}{dt} &= \lambda_r S_r - (\mu_r + \delta_r) I_r.
\end{aligned} \tag{2.1}$$

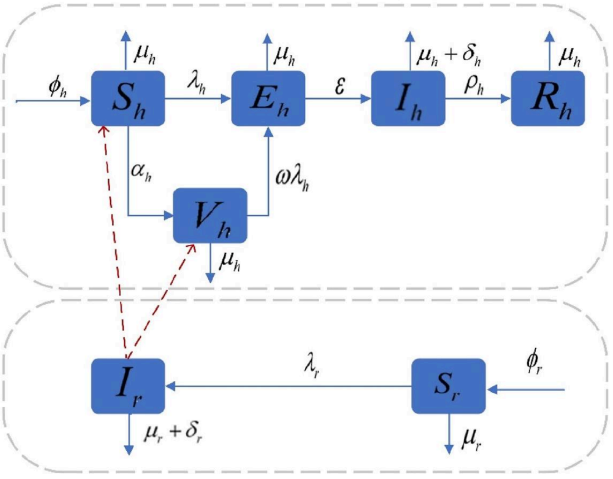


Fig. **1** Flowchart for the Mpox virus transmission The proof is given in Appendix B.

Table 1 Parameters description and parameter values and their sources

Parameters	Description	Parameter value	source
φ_h	Recruitment into susceptible human class	82,123.3	[29]
φ_r	Recruitment into susceptible rodent class	0.112	[29]
μ_h	Natural death rate of humans	0.00968	[32]
μ_r	Natural death rate of rodents	0.00011	[29]
δ_h	Disease-induced death rate of humans	0.07583	Fitting
δ_r	Disease-induced death rate of rodents	0.08732	Fitting
α_h	Human vaccination rate	0.32547	Fitting
ρ_h	Human recovery rate	0.3333	[33]
β_1	Contact rate between susceptible human and infected human	0.08371	Fitting
β_2	Contact rate between susceptible human and infected rodent	0.02941	Fitting
β_3	Contact rate between susceptible rodent and infected rodent	0.05108	Fitting
ε	Progression from exposed to infected class	0.53846	[13]
ω	Correction factor for vaccinee infection rate	0.00648	Fitting

The existence of the equilibrium and the effective reproduction number

In this section, we will demonstrate the existence of the two equilibrium points of system (2.1) and derive the expression for the effective reproduction number. We begin by looking at the disease-free equilibrium.

The disease-free equilibrium point is defined as the point at which all compartments of the system (2.1) are free of disease infection, i.e. $E_h = 0, I_h = 0, R_h = 0, I_r = 0$. Let the right end of all equations in system (2.1) be zero, which gives:

$$S_h^0 = \frac{\varphi_h}{\mu_h + \alpha_h},$$

$$V_h^0 = \frac{\alpha_h \varphi_h}{\mu_h(\mu_h + \alpha_h)},$$

$$S_r^0 = \frac{\varphi_r}{\mu_r},$$

$$E_h^0 = 0, I_h^0 = 0, R_h^0 = 0, I_r^0 = 0.$$

Then the expression for the disease-free equilibrium point of system (2.1) is given by $E^0 = \left(\frac{\varphi_h}{\mu_h+\alpha_h}, \frac{\alpha_h\varphi_h}{\mu_h(\mu_h+\alpha_h)}, 0, 0, \frac{\varphi_r}{\mu_r}, 0\right)^T$.

The effective reproduction number for a disease model is usually determined by the well-known next-generation method. According to the method, we only consider the compartments in which new infections appear, i.e. E_h, I_h, I_r . And the derivation for each of them leads to the transmission matrix F and transitions matrix V:

$$F = \begin{pmatrix} 0 & \frac{\beta_1 S_h + \omega\beta_1 V_h}{N_h} & \frac{\beta_2 S_h + \omega\beta_2 V_h}{N_h} \\ 0 & 0 & 0 \\ 0 & 0 & \frac{\beta_3 S_r}{N_r} \end{pmatrix},$$

$$V = \begin{pmatrix} \varepsilon + \mu_h & 0 & 0 \\ -\varepsilon & \mu_h + \rho_h + \delta_h & 0 \\ 0 & 0 & \mu_r + \delta_r \end{pmatrix},$$

Computing the inverse of V:

$$V^{-1} = \begin{pmatrix} \frac{1}{\varepsilon+\mu_h} & 0 & 0 \\ \frac{\varepsilon}{(\varepsilon+\mu_h)(\mu_h+\rho_h+\delta_h)} & \frac{1}{\mu_h+\rho_h+\delta_h} & 0 \\ 0 & 0 & \frac{1}{\mu_r+\delta_r} \end{pmatrix},$$

Multiplying the matrix F and V^{-1} together, we have.

The proof is given in Appendix E.

$$FV^{-1} = \begin{pmatrix} \frac{\beta_1 S_h+\omega\beta_1 V_h}{N_h} \cdot \frac{\varepsilon}{(\varepsilon+\mu_h)(\mu_h+\rho_h+\delta_h)} & \frac{\beta_1 S_h+\omega\beta_1 V_h}{N_h} \cdot \frac{1}{\mu_h+\rho_h+\delta_h} & \frac{\beta_1 S_h+\omega\beta_1 V_h}{N_h} \cdot \frac{1}{\mu_r+\delta_r} \\ 0 & 0 & 0 \\ 0 & 0 & \frac{\beta_3 S_r}{N_r} \cdot \frac{1}{\mu_r+\delta_r} \end{pmatrix},$$

Substituting the expression of E^0 , we have

Stability of the endemic equilibrium

This section will prove the global stability of the endemic

$$\Re_0 = \rho\left(\mathrm{FV}^{-1}\right) = \max\left\{ \frac{\varepsilon\beta_1(\mu_h+\omega\alpha_h)}{(\mu_h+\alpha_h)(\varepsilon+\mu_h)(\mu_h+\rho_h+\delta_h)}, \frac{\beta_3}{\mu_r+\delta_r} \right\} \tag{3.8}$$

Where $\Re_{h_0} = \frac{\varepsilon\beta_1(\mu_h+\omega\alpha_h)}{(\mu_h+\alpha_h)(\varepsilon+\mu_h)(\mu_h+\rho_h+\delta_h)}, \Re_{r_0} = \frac{\beta_3}{\mu_r+\delta_r}.$

Theorem 3.3. *When* $\Re_0 > 1$ *, the system (2.1) has a unique endemic equilibrium point* E^* *.*

The proof is given in Appendix C.

Stability analysis

Stability of disease-free equilibrium

This section will prove the local asymptotic stability and global stability of the disease-free equilibrium point E^0 obtained in the previous section.

Theorem 3.4. *When* $\Re_0 < 1$ *, the disease-free equilibrium* E^0 *of the system (2.1) is locally asymptotically stable, when* $\Re_0 > 1$, E^0 *is unstable.*

The proof is given in Appendix D.

The global stability of the disease-free equilibrium E^0 is proved below.

Theorem 3.5. *When* $\Re_0 < 1$ *, the disease-free equilibrium* E^0 *of the system (2.1) is globally stable under special circumstances, that is, when vaccination is not considered* ($\alpha_h = 0$).

equilibrium point of the system (2.1). According to theorem 3.3, when $\Re_0 \leq 1$, there is no endemic equilibrium point in system (2.1). When $\Re_0 > 1$, the only endemic equilibrium exists. The following proves that when $\Re_0 > 1$, the endemic equilibrium point of system (2.1) is globally stable by constructing the Lyapunov function.

Theorem 3.6. *When $\Re_0 > 1$, the endemic equilibrium of the system (2.1) is globally asymptotically stable under special circumstances, that is, when natural death is not considered* ($\delta_h = \delta_r = 0$) .

The proof is given in Appendix F.

Parameter estimation and sensitivity analysis

Parameter estimation

The main purpose of this section is to complete the estimation of the parameter values contained in the system (2.1), some of which are determined by consulting relevant references, but the key parameter values will be fitted according to actual statistical data.

The Mpox outbreak has affected different countries around the world to varying degrees. Figure 2 shows the

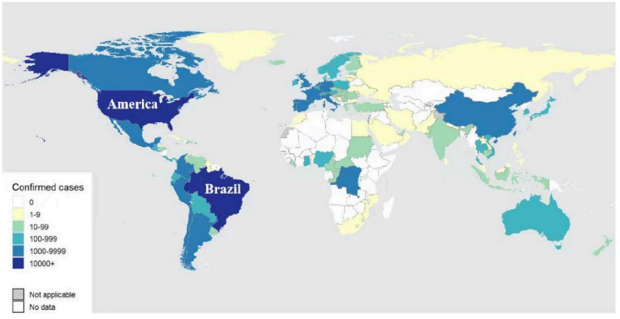


Fig. 2 Global distribution of Mpox cases (https://worldhealthorg.shinyapps.io/mpx_global/#23_Maps) global distribution of Mpox cases as of 29 February 2024. As can be observed from the figure, countries such as the United States and Brazil are the regions most affected by the epidemic, and the number of confirmed cases reported is particularly prominent.

In this paper, using the world health organization (WHO) reported data for weekly confirmed Mpox infection cases from 2022.6.1 to 2023.11.28 in the United States (https://worldhealthorg.shinyapps.io/mpx_global/), the nonlinear least squares fitting method [34] is used to minimize the error between the simulated data and the actual data to obtain the optimal parameter value of the model. The data from the first 55 weeks are used as the training set and the data from the last 20 weeks are used as the test set to verify the validity of the obtained parameter values. The loss function used is as follows:

$$L = \sum_{i=1}^{\tau} \left(\hat{I}_h(i) - I_h(i) \right)^2.$$

Where, $\hat{I}_h$ represents the number of infection cases obtained by fitting, I_h represents the actual number of infection cases reported, and τ represents the length of data. This is a minimization problem, the expected parameter values θ can minimize the error shown in the above formula, so that the model is consistent with the actual situation to the greatest extent, that is:

$$\theta = \arg\min_{\theta} \sum_{i=1}^{\tau} \left(\hat{I}_h(i,\theta) - I_h(i)\right)^2.$$

During the fitting process, the initial values of the state variables contained in the system (2.1) are set. For the initial value setting of human susceptible persons, this paper referred to literature [35], that is, age distribution was considered. Mpox virus belongs to the poxvirus family and belongs to the orthopoxvirus genus [12] with smallpox virus. Their antigenic properties are basically the same and they have cross-immunity with each other [36, 37]. Smallpox vaccination has certain protective effects on Mpox virus infection. According to reports, people vaccinated against smallpox are 1/5.2 times more likely to be infected with Mpox virus than those who are not vaccinated. Since smallpox vaccination was stopped after the global eradication of smallpox in the 1980 s, this paper assumes that people aged 40 and above were vaccinated against smallpox, while people under 40 were not vaccinated [38, 39]. Accordingly, the initial value of human susceptibility S_h is set as:

S_{h0}= Population under the age of 40 $+\frac{1}{5.2}\times$ Population over 40 years old.

By looking at the United Nations statistics division database (https://unstats.un.org/UNSDWebsite/), you can get all ages population data in 2022. The United States to human susceptible people S_h and vaccinees V_h initial value is set to:

$$S_{h0} = 202{,}651{,}928,\ V_{h0} = 161{,}739{,}350.$$

The initial values of human infected persons I_h and human recovered persons R_h are respectively taken from the actual data of the United States on June 1, 2022. If the initial values of human latent persons E_h are $E_{h0} = 3 * I_{h0}$, they are set as follows:

$$E_{h0} = 75, I_{h0} = 25, R_{h0} = 0.$$

The initial values of animal groups, including susceptible animals S_r and infected animals I_r , are selected with reference to literature [29], that is, they are respectively set as:

$$S_{r0} = 90{,}000, I_{r0} = 10.$$

The model was numerically simulated with the above initial values, and the fitting and obtained parameter values were shown in Fig. 3 and Table 1, respectively.

Figure 3 shows the fitting results for the weekly number of Mpox cases in the United States between June 1, 2022 and November 28, 2023, where the blue dots represent the actual data and the blue lines represent the model fit values. After about 25 weeks, it was observed that there was a significant agreement between the fitted data and the actual data, and the fitting effect was very good.

The fitted specific parameter values (given in Table 1) were substituted back into system (2.1) to predict the Mpox epidemic in the United States, and the results were shown in Fig. 4.

It can be seen from Fig. 4 that the predicted results of the specific parameters given in Table 1 are almost consistent with the actual situation, and it can be seen that the system parameter values fitted by the least square nonlinear regression method in the previous paper are effective. In addition, it can be seen from the figure that the Mpox epidemic in the United States will continue to exist in the short term. Therefore, it is necessary to further understand the transmission characteristics of Mpox virus and pay attention to the prevention and control of Mpox epidemic.

Sensitivity analysis

Understanding the key factors affecting the development of Mpox epidemic is a prerequisite for the prevention and control of Mpox. Therefore, in this section, the key parameters affecting the transmission of Mpox virus in the analysis model (2.1) will be discussed, and reasonable prevention and control measures will be proposed.

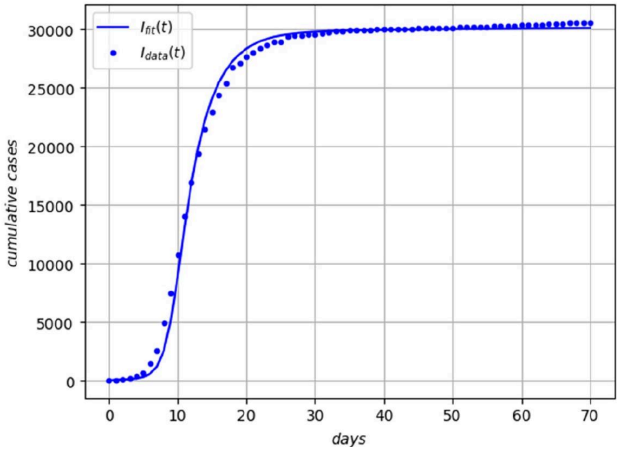


Fig. 3 Model fitting results

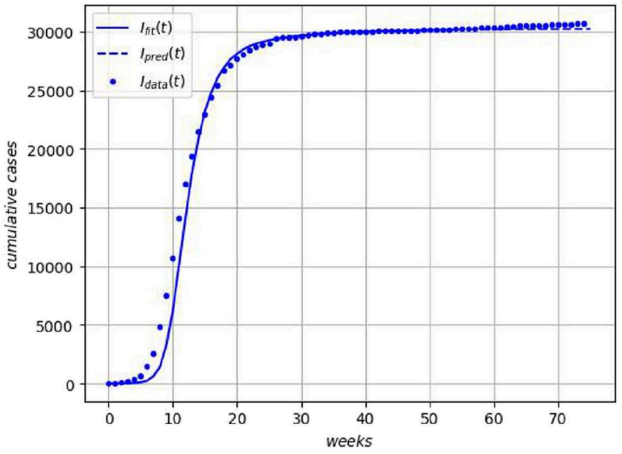


Fig. 4 Model prediction results

Local sensitivity analysis

The normalized forward sensitivity index [40] is one of the important tools used to analyze the factors that have a significant impact on the spread and control of epidemics and can measure the relationship between the relative change of parameters and the relative change of the effective reproduction number. In this section, the normalized sensitivity analysis of the parameters in the model (2.1) was carried out to explore the degree of influence of each parameter on Mpox transmission.

Definition 4.1: Defines the normalized forward sensitivity index of measurement $\Re_0$ with respect to relative changes in model parameters as:

$$Y_x^{\Re} = \frac{x}{\Re_0} \times \frac{\partial \Re_0}{\partial x}. \tag{4.1}$$

Where x is the corresponding model parameter and $\Re_0$ is the effective reproduction number.

By using formula (4.1), the partial derivative of the expression of the effective reproduction number $\mathfrak{R}_0$ with respect to each parameter can be obtained, and the sensitivity index of the corresponding parameter can be calculated by substituting the specific parameter values given in Table 1, as shown in Table 2. When the value is positive, it means that the corresponding parameter is proportional to the effective reproduction number, that is, when the parameter value increases, the $\Re_0$ value increases. Conversely, it is inversely proportional, that is, when the parameter value decreases, the $\Re_0$ value increases.

According to Table 2, the person-to-person contact rate (β_1), the reciprocal of the human incubation period (ε) and the correction factor for the infection rate of vaccinators (ω) are proportional to the effective reproduction number, while the case fatality rate (δ_h), recovery rate (ρ_h) and vaccination rate (α_h) are inversely proportional to the effective reproduction number. Among them, the most significant influence on the effective reproduction number is the person-to-person contact rate, followed by the recovery rate, vaccination rate and the

infection correction factor of vaccinators. This suggests that when formulating prevention and control measures to contain the development of Mpox virus, in addition to timely isolation of Mpox infection cases and vaccination of highrisk populations, it is also important to pay attention to the research and development of Mpox specific drugs and vaccines to improve the treatment level of Mpox virus and the protective efficacy of vaccines against Mpox virus.

Global sensitivity analysis

In the previous section, several parameters that would have an important impact on the effective reproduction number were preliminatively obtained by using the normalized forward sensitivity index. However, since they measured local sensitivity, that is, the output changes were observed by perturbing parameters at a certain point, it was impossible to know the impact of large changes on the effective reproduction number. In this section, the Latin Hypercube sampling method (LHS) was used to analyze the global sensitivity of the model parameters by solving the partial rank correlation coefficient (PRCC), and the relationship between each parameter and the effective reproduction number $\Re_0$ was given more intuitively, as shown in Fig. 5.

Similar to the conclusions in the previous section, it can be seen from Fig. 5 that person-to-person contact rate (β_1) , infection correction factor of vaccinators (ω), vaccination rate (α_h) and disease recovery rate (ρ_h) have significant effects on the effective reproduction number $\Re_0$. Among them, the person-to-person contact rate β_1 and the infection rate correction factor of vaccinators ω are positively correlated with the effective reproduction number $\Re_0$, that is, the contact between people can be reduced by taking home, isolation and other prevention and control measures, and more human and financial

Table 2 Local sensitivity indicators

Parameter symbol	About $\Re_{h_0}$ the sensitivity indicators
δ_h	− 0.1812
ρ_h	− 0.7957
α_h	− 0.6146
β_1	1.0000
ε	0.0177
ω	0.3565

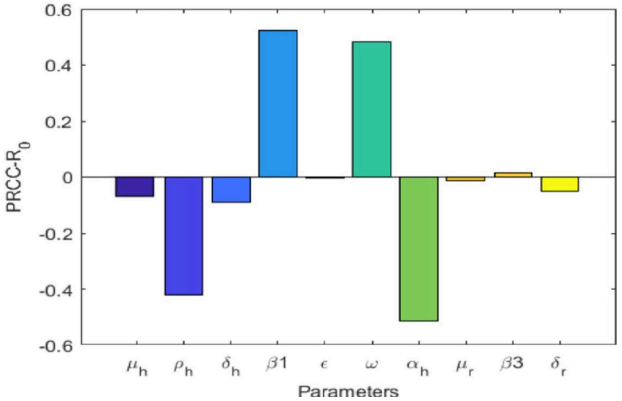


Fig. 5 Global sensitivity indicators resources can be invested in the research and development of vaccines against Mpox virus, so as to further improve its protective effect against Mpox virus and reduce the effective reproduction number. The vaccine coverage rate α_h and recovery rate ρ_h are negatively correlated with the effective reproduction number $\Re_0$, that is, the effective reproduction number can be reduced by improving the Mpox vaccine coverage rate and the treatment level of Mpox virus, so as to achieve the effect of controlling the Mpox epidemic.

From the observation of Fig. 5, it can be further learned that the protective efficacy of Mpox vaccine is also one of the important factors affecting the development of Mpox epidemic, which suggests that in the prevention and control of Mpox epidemic in the future, more attention should be paid to the protective efficacy of the vaccine, more efforts should be made to develop the Mpox virus vaccine, and more human and financial resources should be invested to develop the Mpox vaccine with stronger protective efficacy as soon as possible.

Optimal control

As can be seen from the above, the person-to-person contact rate (β_1) , recovery rate (ρ_h) , vaccination rate (α_h) and the infection rate correction factor (ω) of vaccinators have significant effects on epidemic prevention (ii) $u_2(t)$: This variable is used to increase the coverage of Mpox vaccine (α_h) ,

(iii) $u_3(t)$: This variable is used to improve the protective efficacy of the Mpox vaccine, i.e. to reduce the infection rate of the vaccinated person (ω),

(iv) $u_4(t)$: This variable is used to control the recovery rate of Mpox infection (ρ_h) , that is, to improve the level of treatment for Mpox infection.

By using the above control variables and on the basis of the original model (2.1), a new optimal control model is developed in this section, as follows:

$$\begin{aligned}
\frac{\mathrm{d}S_h}{\mathrm{d}t} &= \varphi_h - \lambda_h(1-u_1)S_h - u_2S_h - \mu_hS_h,\\
\frac{dV_h}{dt} &= u_2S_h - \omega\lambda_h(1-u_3)V_h - \mu_hV_h,\\
\frac{dE_h}{dt} &= \lambda_h(1-u_1)S_h + \omega\lambda_h(1-u_3)V_h - (\varepsilon+\mu_h)E_h,\\
\frac{dI_h}{dt} &= \varepsilon E_h - (\rho_h+u_4)I_h - (\mu_h+\delta_h)I_h,\\
\frac{dR_h}{dt} &= (\rho_h+u_4)I_h - \mu_hR_h,\\
\frac{dS_r}{dt} &= \varphi_r - \lambda_rS_r - \mu_rS_r,\\
\frac{dI_r}{dt} &= \lambda_rS_r - (\mu_r+\delta_r)I_r.
\end{aligned} \tag{5.1}$$

Since the purpose of control is not only to reduce the number of latent and infected Mpox patients, but also to minimize the cost of prevention and control measures, the corresponding objective function is set as follows:

$$J(u_1,u_2,u_3,u_4) = \int_0^T \left((A_1E_h + A_2I_h) + \frac{1}{2}(b_1u_1^2 + b_2u_2^2 + b_3u_3^2 + b_4u_4^2)\right)dt.$$

and control, but in real life, the impact of economic and other social factors should also be considered. For example, although long-term home isolation and other prevention and control measures can effectively curb the spread of related

diseases, they will also greatly affect residents'daily life and social and economic development. Therefore, this section applies the optimal control theory to the original model, takes the above several parameters as control variables, establishes a new optimal control model, and seeks the most appropriate prevention and control measures under the optimal control framework, so as to effectively control the number of Mpox virus incubators and infected persons, and minimize the social epidemic prevention cost.

Consider the following four time-dependent control variables:

(i) $u_1(t)$:This variable is used to reduce contact between susceptible persons and those infected with Mpox by taking measures such as personal protection(β_1,β_2) , The symbols A_1 and A_2 are the balance factors of latent and infected Mpox patients respectively. b_1,b_2,b_3 and b_4 are the weight coefficients of the control variables u_1, u_2 , u_3 and u_4 respectively, that is, the corresponding cost factors; T stands for time step. The main goal of this section is to expect to find an optimal control that minimizes a given objective function, i.e

$$\overline{u}_i(t) \ i = 1, 2, 3, 4,$$

we have

$$J(\overline{u}_1, \overline{u}_2, \overline{u}_3, \overline{u}_4) = \min_{\Xi} \{J(u_1, u_2, u_3, u_4)\},$$

where is the Lebesgue measurable set:

$$\Xi = \{(u_1, u_2, u_3, u_4) : [0, T] \to [0, 1]\}.$$

Before dealing with the optimal control problem, we first prove the existence and uniqueness of the optimal problem, and the relevant proposition and its proof is in Appendix G.

According to the Pontryagin maximum principle [41], the optimal control problem can be transformed into the minimum problem of finding the Hamilton function. By introducing adjoint variables $\lambda_i (i = 1, \ldots, 7)$, the Hamilton function corresponding to the above optimal control model (5.1) can be easily obtained as follows:

$$\begin{aligned} H = {} & A_1 E_h + A_2 I_h + \tfrac{1}{2}(b_1 u_1^2 + b_2 u_2^2 + b_3 u_3^2 + b_4 u_4^2) \\ & + \lambda_1(\varphi_h - \lambda_h(1 - u_1)S_h - u_2 S_h - \mu_u S_h) \\ & + \lambda_2(u_2 S_h - \omega\lambda_h(1 - u_3)V_h - \mu_h V_h) \\ & + \lambda_3(\lambda_h(1 - u_1)S_h + \omega\lambda_h(1 - u_3)V_h - (\varepsilon + \mu_h)E_h) \\ & + \lambda_4(\varepsilon E_h - (\rho_h + u_4)I_h - (\mu_h + \delta_h)I_h) \\ & + \lambda_5((\rho_h + u_4)I_h - \mu_h R_h) \\ & + \lambda_6(\varphi_r - \lambda_r S_r - \mu_r S_r) \\ & + \lambda_7(\lambda_r S_r - (\mu_r + \delta_r)I_r). \end{aligned} \tag{5.2}$$

By taking the derivative of the Hamilton function (5.2) with respect to the state variable, we can see that the adjoint variable $\lambda_i (i = 1, \ldots, 7)$ satisfies the following adjoint system: Among them, the adjoint equation also needs to meet the following transversal conditions:

$$\lambda_i(T) = 0, i = 1, \ldots, 7$$

In addition, the expression of the optimal control solution $(\overline{u}_1, \overline{u}_2, \overline{u}_3, \overline{u}_4)$ corresponding to the above minimization problem can be obtained by solving $\frac{\partial H(t,u_j,\lambda_i)}{\partial u_j} = 0$:

$$\begin{aligned} \overline{u}_1 &= \min\left\{1, \max(0, \frac{(\lambda_3-\lambda_1)(\beta_1 I_h+\beta_2 I_r)}{b_1 N_h})\right\}, \\ \overline{u}_2 &= \min\left\{1, \max(0, \frac{(\lambda_1-\lambda_2)S_h}{b_2})\right\}, \\ \overline{u}_3 &= \min\left\{1, \max(0, \frac{(\lambda_3-\lambda_2)(\omega\beta_1 I_h+\omega\beta_2 I_r)V_h}{b_3 N_h})\right\}, \\ \overline{u}_4 &= \min\left\{1, \max(0, \frac{(\lambda_4-\lambda_5)I_h}{b_4})\right\}. \end{aligned} \tag{5.4}$$

$$\begin{aligned} \frac{d\lambda_1}{dt} &= u_2(\lambda_1-\lambda_2)+(\lambda_1-\lambda_3)(1-u_1)\left(\frac{\beta_1 I_h+\beta_2 I_r}{N_h}\right)+(\lambda_3-\lambda_1)\left(\frac{\beta_1 I_h+\beta_2 I_r}{N_h^2}\right)(1-u_1)S_h \\ &+(\lambda_3-\lambda_2)\left(\frac{\omega\beta_1 I_h+\omega\beta_2 I_r}{N_h^2}\right)(1-u_3)V_h+\lambda_1\mu_h, \\ \frac{d\lambda_2}{dt} &= (\lambda_3-\lambda_1)\left(\frac{\beta_1 I_h+\beta_2 I_r}{N_h^2}\right)(1-u_1)S_h+(\lambda_2-\lambda_3)\left(\frac{\omega\beta_1 I_h+\omega\beta_2 I_r}{N_h}\right)(1-u_3) \\ &+(\lambda_3-\lambda_2)\left(\frac{\omega\beta_1 I_h+\omega\beta_2 I_r}{N_h^2}\right)(1-u_3)V_h+\lambda_2\mu_h, \\ \frac{d\lambda_3}{dt} &= -A_1+\epsilon(\lambda_3-\lambda_4)+(\lambda_3-\lambda_4)\left(\frac{\beta_1 S_h+\beta_2 S_r}{N_h^2}\right)(1-u_1)S_h \\ &+(\lambda_3-\lambda_2)\left(\frac{\omega\beta_1 S_h+\omega\beta_2 S_r}{N_h^2}\right)(1-u_3)V_h+\lambda_3\mu_h, \\ \frac{d\lambda_4}{dt} &= -A_2+(\rho_h+u_4)(\lambda_4-\lambda_5)+(\lambda_1-\lambda_3)\left(\frac{\beta_1 S_h}{N_h}\right)(1-u_1) \\ &+(\lambda_1-\lambda_3)\left(\frac{\beta_1 I_h+\beta_2 I_r}{N_h^2}\right)(1-u_1)S_h+(\lambda_2-\lambda_3)\left(\frac{\omega\beta_1 V_h}{N_h}\right)(1-u_3) \\ &+(\lambda_3-\lambda_2)\left(\frac{\omega\beta_1 I_h+\omega\beta_2 I_r}{N_h^2}\right)(1-u_3)V_h+\lambda_4(\mu_h+\delta_h), \\ \frac{d\lambda_5}{dt} &= (\lambda_3-\lambda_1)\left(\frac{\beta_1 I_h+\beta_2 I_r}{N_h^2}\right)(1-u_1)S_h+(\lambda_3-\lambda_2)\left(\frac{\omega\beta_1 I_h+\omega\beta_2 I_r}{N_h^2}\right)(1-u_3)V_h+\lambda_5\mu_h \\ \frac{d\lambda_6}{dt} &= (\lambda_6-\lambda_7)\left(\frac{\beta_2 I_r}{N_r}\right)+(\lambda_7-\lambda_6)\left(\frac{\beta_2 S_r I_r}{N_r^2}\right)+\lambda_6\mu_r, \\ \frac{d\lambda_7}{dt} &= (\lambda_1-\lambda_3)\left(\frac{\beta_2 S_h}{N_h}\right)(1-u_1)+(\lambda_2-\lambda_3)\left(\frac{\omega\beta_2 V_h}{N_h}\right)(1-u_3)+(\lambda_6-\lambda_7)\frac{\beta_2 S_r}{N_r} \\ &+(\lambda_7-\lambda_6)\frac{\beta_2 S_r I_r}{N_r^2}+\lambda_7(\mu_r+\delta_r). \end{aligned} \tag{5.3}$$

Numerical simulation

Global stability of equilibrium

Theorems 3.5 and 3.6 show that disease-free equilibria E^0 are globally asymptotically stable without taking into account vaccination, and endemic equilibria E^* are globally stable without taking into account natural death of humans and animals. Therefore, different initial values were selected to draw the track plots of human susceptible persons, vaccinators and infected persons for numerical simulation. For the disease-free equilibrium E^0 , the value of the parameter is $\varphi_h =$ 11,731.91, $\varphi_r =$ 0.016, $\mu_h =$ 0.000036, $\mu_r =$ 0.000016, $\beta_1 =$ 0.2701, $\beta_2 =$ 0.2508, $\beta_3 =$ 0.002461, $\delta_h =$ 0.0011, $\delta_r = 0.0153$, $\varepsilon = 0.0486$, $\alpha_h =$ 0, $\rho_h =$ 0.467, $\omega = 0.265$. At this time, the effective reproduction number $\Re_0 < 1$; for the endemic equilibria E^* , the value of the parameter is $\varphi_h =$ 11,731.91, $\varphi_r =$ 0.016, $\mu_h =$ 0.000036, $\mu_r =$ 0.000016, $\beta_1 =$ 1.8701, $\beta_2 =$ 0.2508, $\beta_3 =$ 0.2461, $\delta_h =$ 0, $\delta_r = 0$, $\varepsilon = 0.0486$, $\alpha_h =$ 0.2393, $\rho_h =$ 0.467, $\omega =$ 0.265. At this time, the effective reproduction number $\Re_0 > 1$.

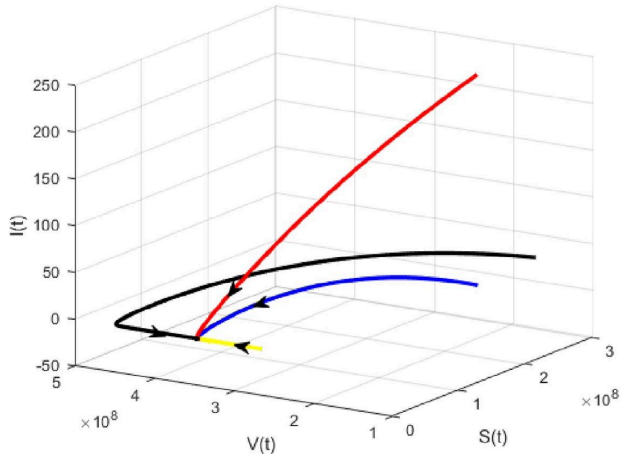


Fig. 6 Global stability of the disease-free equilibrium

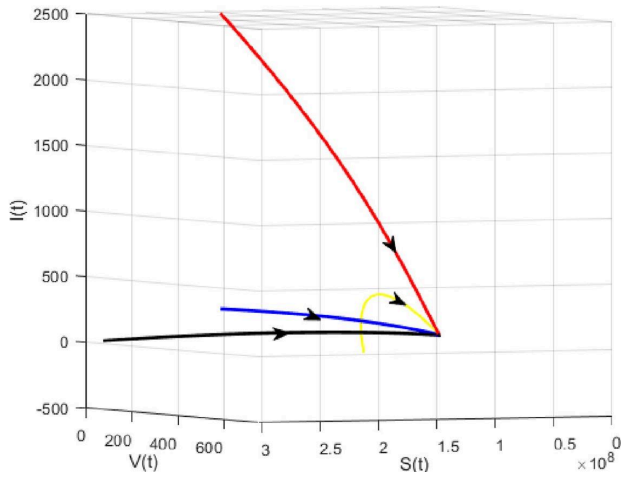


Fig. 7 Global stability of the endemic equilibrium The simulation results are shown in Figs. 6 and 7. It can be seen from the figure that tracks with different initial values will eventually coincide at the same point, indicating that the disease-free equilibrium and the endemic equilibrium of model (2.1) have global stability, thus verifying the accuracy of the analysis results in Sect. 3.

Optimal control

In this section, we conduct detailed numerical simulations of the established optimal control model (5.1) and the model without control (2.1) by using the forward–backward scanning method and the fourth-order Runge–Kutta method to explore the impact and effect of the proposed control intervention on the development of Mpox disease in the United States. The main process of the algorithm is as follows:

This paper simulates various scenarios by combining different control variables. Considering that control variable u_3 represents the implementation of measures to improve the protective efficacy of Mpox vaccine, and this measure has practical significance only after the implementation of vaccination, the implementation of only measures u_3 to improve the protective efficacy of vaccine is not considered when implementing a single control variable; And when controlling variables are combined, measures to improve vaccine protective efficacy must be implemented at the same time as measures u_2 ,

which represents an increase in vaccination rates. Based on this, the following control schemes are mainly considered (Table 3):

In the numerical simulation, the parameter values given in Table 1 are used as input, and the remaining weight coefficient values are set as $A_1 = 100, A_2 = 200$, $b_1 = 30$, $b_2 = 20, b_3 = 20, b_4 = 10$. Set the simulation duration to 300 days. In the following, model simulation was performed for the four control schemes proposed in Table 3, and their dynamic changes on model state variables were observed to analyze the impact of these control measures on the development of Mpox epidemic in the United States. The specific simulation results are shown in Fig. 8 and Appendix H, (see Appendix H, Figure A1, A2, A3, A4, A5, A6, A7, A8), where the red line segment indicates that the corresponding control measures have been implemented, and the blue line segment indicates that no control measures have been implemented.

For the fourth scenario, that is, taking into account the implementation of personal protection measures u_1 , vaccination of Mpox vaccine u_2 , and measures to improve the protective efficacy of the vaccine u_3 and the level of disease treatment u_4 . Figure 8 gives specific simulation results. It can be seen that, compared with the control plan proposed above, the simultaneous implementation of these four measures has a significant effect on reducing the number of latent and infected persons, and can also increase the number of vaccinated persons. As can be seen from the corresponding control curve, the personal protection measures remained at 0.8 intensity for the first 73 days, and then began to decline to the lower limit at the end of the simulation. Measures to increase Mpox vaccination remained at maximum intensity for the first 6 days, then gradually decreased to the lower limit at about day 200 until the end of the simulation period. Measures to enhance the protective efficacy of Mpox vaccine remained at maximum intensity for about 40 days, then gradually decreased to the lower limit at about 100 days until the end of the simulation period. The measures to increase the therapeutic level of Mpox virus remained at the highest intensity for about 118 days before gradually decreasing to the lower limit at about 223 days until the end of the simulation period.

According to the above simulation results, it can be seen that the implementation of the four prevention and control measures proposed in this paper is the most effective to curb the development of the Mpox epidemic in the United States, considering both the reduction of the number of latent Mpox patients and infected persons and the cost of implementing corresponding prevention and control measures.

Table 3 Control schemes

Scheme	Description	Implementation measure
Scheme One	Single control variable	$u_i \neq 0, i = 1, 2, 4$
Scheme Two	Dual control variable	$u_1, u_2 \neq 0$ or $u_1, u_4 \neq 0$ or $u_2, u_3 \neq 0$
SchemeThree	Triple control variable	$u_1, u_2, u_3 \neq 0$ or $u_2, u_3, u_4 \neq 0$
Scheme Four	All control variables	$u_1, u_2, u_3, u_4 \neq 0$

Conclusion

In this paper, the Mpox epidemic was taken as the research object, after analyzing the transmission characteristics and research status of Mpox virus, the protective efficacy of Mpox vaccine was considered for the first time, and a new Mpox dynamic model was proposed to study the transmission dynamics of Mpox virus by introducing the virus exposure correction factor of vaccinators. According to the transmission characteristics of Mpox virus, the model is composed of two subsystems, human and animal, which are specifically divided into seven chambers. The human subsystem includes five chambers, which are susceptible (S_h) , vaccinated (V_h) , latent (E_h) , infected (I_h) , and recovered (R_h);The non-human subsystem includes susceptible (S_r) and infected (I_r) . Firstly, the model is qualitatively analyzed, which proves the positive and boundedness of model, and the reasonable validity of the model is guaranteed. The expression of the effective reproduction number of the model was obtained by using the next generation matrix method, and the existence and stability of the disease-free equilibrium and the endemic equilibrium of the model were proved. The stability of the two equilibrium was verified by numerical simulation, which fully demonstrated the accuracy of the proof results.

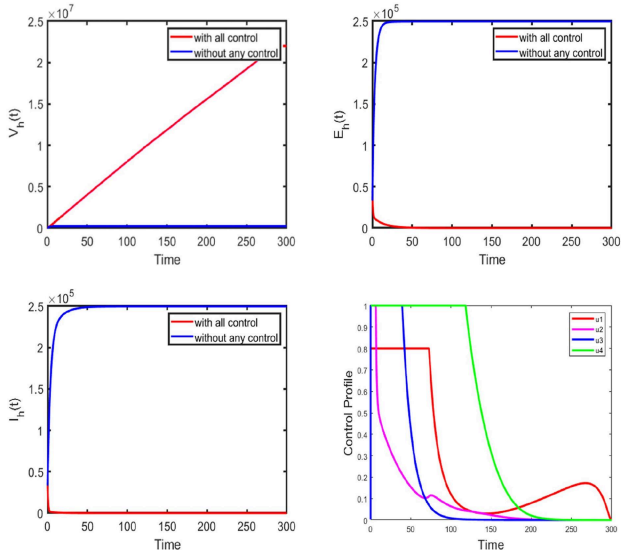


Fig. 8 The effect of implementing all control measures u_1, u_2, u_3, u_4 on changes in the number of vaccinated populations, exposed and infected populations, and the corresponding time series of optimized control functions

The model parameter values are obtained by fitting. As the United States is one of the most seriously affected countries in the global Mpox outbreak, the optimal parameter values were obtained by using the Mpox infection data of the United States from 2022 to 2023 released by the WHO and the nonlinear least square method was used to fit the model. The prediction results indicated that the Mpox epidemic in the United States would persist in the short term. The sensitivity of model parameters was analyzed, and the sensitivity index of model parameters relative to effective reproduction number was obtained by using normalized forward sensitivity index and Latin hypercube sampling method. The influence of model parameters on Mpox virus transmission was quantified, and the key factors affecting the development of Mpox epidemic in the United

States were obtained. It is important to improve the effectiveness of Mpox vaccine to curb the spread of Mpox.

Finally, based on the optimal control theory, four interventions are proposed, which are: reducing the contact between susceptible people and monkeypox infected people, increasing the coverage of monkeypox vaccine, improving the protective efficacy of monkeypox vaccine, and improving the treatment level of monkeypox infection. A new optimal control problem is constructed in order to obtain the most effective control strategy at the least cost. This paper puts forward single, double, three and four prevention measures. The effects of control strategies on monkeypox outbreak were simulated and analyzed. It is concluded that the implementation of four control measures, that is, the implementation of four control measures at the same time, is the most reasonable and effective way to curb the development of monkeypox epidemic in the United States. Therefore, for effective implementation of monkeypox control measures, policy makers and public health authorities should provide the following advice: Conduct public awareness campaigns to educate individuals about the importance of personal protection, such as wearing masks, practicing good hygiene and social distancing. Strengthen isolation programmes for those infected to minimize community transmission. Prioritize equitable distribution of the Mpox vaccine to ensure access to at-risk populations and underserved communities. Invest in research and development to improve the protective efficacy of the Mpox vaccine. Ensure proper storage, handling and management of vaccines to maintain their effectiveness. Strengthen health care infrastructure to provide timely and effective treatment for measles infection.

The article has the following limitations: Oversimplification of rodent population: only two types of susceptibility and infection are distinguished, without considering the complexity of rodent transmission (such as incubation period and secondary infection), which may lead to underestimation or overestimation of the risk of rodents as sources of infection. Spatial heterogeneity is not considered: The model assumes that all humans and rodents are evenly mixed in the same area, ignoring geographic isolation, differences in population behavior, or unequal distribution of rodent habitats. Social interventions are not included: the impact of dynamic changes in vaccination coverage, isolation policies or public health interventions on transmission chains is not taken into account. In addition, the control efforts considered in this study focused only on human populations, and no specific studies on animal populations were conducted in this study. Reasonable control measures targeting animal host populations can be considered as potential interventions to prevent monkeypox virus transmission. We will further improve the depth and breadth of our model and research, which is the direction of our future efforts.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12879-025-10873-y.

Supplementary Material 1.

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Not applicable.

Authors'contributions

MQ: Conceptualization, Methodology, Software, Data Curation, Visualization, Writing-Original Draft. WL: Conceptualization, Supervision, Resources, WritingReviewing and Editing, Funding acquisition. DL: Visualization, Data Curation, Visualization. ZH: Methodology, Software, Data Curation. SH: Resources, Writing-Reviewing and Editing, Funding acquisition.

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Data availability

The data that support the findings of this study are available from World Health Organization (https://worldhealthorg.shinyapps.io/mpx_global/.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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